FinRL-Trading: Machine-Learning Stock Selection and a Price-Covered Top-20 Portfolio

STAT GR5398 | Assignment 1 | Exploratory research report | 11 October 2026

Study window: 1 May 2020 to 30 September 2025. Benchmark: SPY adjusted closing prices. Scope: eleven equity sectors; sector-level machine-learning prediction followed by equal-weight portfolio construction.

Executive summary

This study evaluates a fundamental-feature stock-selection workflow using Random Forest, LightGBM, and XGBoost across eleven sector datasets. Daily candidate predictions are ranked; the final Price-Covered Top 20 strategy selects the twenty highest-ranked eligible securities available in the supplied historical price panel, then uses the preceding signal date's weights for adjusted close-to-close returns. A simplified transaction-cost deduction is included.

Over the aligned evaluation window, the restricted-universe strategy accumulated 194.76% versus 148.07% for SPY, an arithmetic cumulative-return gap of 46.69 percentage points. However, its Sharpe ratio was lower (0.93 versus 1.05) and maximum drawdown was substantially deeper (-38.95% versus -24.50%). These findings are exploratory, not fully point-in-time or independently out-of-sample validated.

1. Data and research design

The local workflow uses prepared fundamental-ratio files, sector-specific inputs, and daily equity prices from the course project files. The executed notebook loaded precomputed `prepared/` data and a saved `stock_selected.csv`; it did not rerun preprocessing or retrain the models during the final evaluation. The prediction file contains 171,013 stock-date records, 480 distinct predicted tickers, and signal dates from 1 May 2020 to 25 November 2025. The tested return window ends on 30 September 2025.

The feature collection contains accounting and valuation measures such as EPS, BPS, profitability, liquidity, leverage, and cash-flow ratios. Source fields include `date` and `reportdate`; their point-in-time interpretation has not been established. A historical-constituents file exists in the project, but its full integration into the final eligible universe and treatment of delistings were not verified.

2. Model and portfolio methodology

2.1 Sector-based stock selection

The stock-selection runner processes sectors 10, 15, ..., 60, uses sector prediction outputs generated with Random Forest, LightGBM and XGBoost, and retains the highest predicted-return quartile of combined candidates on each signal date. Model hyperparameter settings are maintained in the accompanying `ml_model.py`; this report does not claim an independently audited cross-validation or walk-forward protocol.

2.2 Restricted price coverage and Top-20 weighting

The original prediction universe includes 67 tickers absent as entire columns from `prepared/daily.csv`; these include both historical corporate-action cases and other missing securities. A separate SQLite price table did not provide records for those 67 identifiers. The final strategy therefore redefines eligibility before ranking: among predicted tickers represented in the supplied price panel, it takes up to 20 names per signal date and gives them equal weights. This is not a repair of the unrestricted Top-20 portfolio and may introduce coverage/survivorship bias.

2.3 Return accounting

Stock adjusted closes are calculated as `prccd / ajexdi`, consistent with the notebook code. Daily close-to-close changes are combined with weights lagged by one row of common trading dates. A fee rate of 0.001 (10 basis points) is applied to the sum of absolute day-to-day changes in the lagged target weights. SPY uses Yahoo Finance adjusted closing prices; both return streams are aligned by trading date. The final run reported zero days with missing returns among active positions, but this does not establish correct delisting payoffs or equal adjustment conventions between data vendors.

3. Main performance results

Metric	Price-Covered Top 20	SPY
Cumulative return	194.76%	148.07%
Annualized return	22.16%	18.32%
Annualized volatility	24.74%	17.55%
Sharpe ratio (risk-free rate = 0)	0.93	1.05
Maximum drawdown	-38.95%	-24.50%

The candidate strategy achieved higher nominal historical growth, but with markedly greater volatility and a larger peak-to-trough loss. Its lower Sharpe ratio means that higher cumulative return should not be interpreted as stronger risk-adjusted performance.

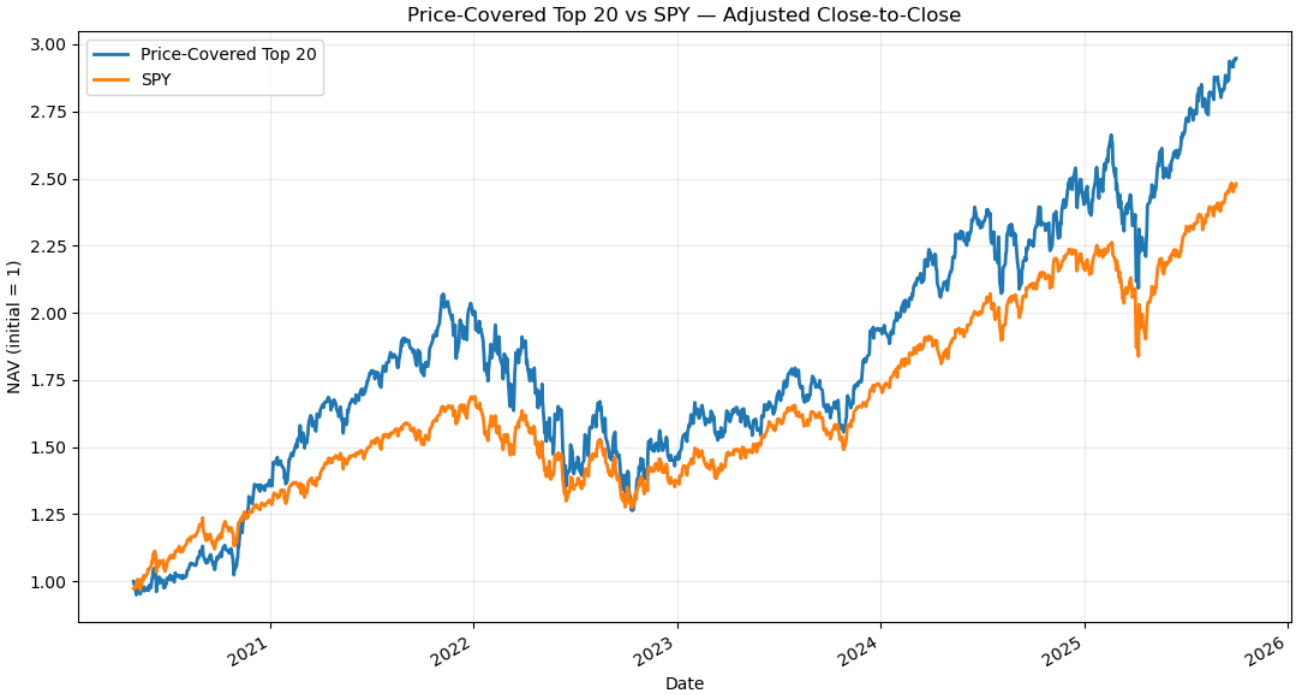


Figure 1. Aligned adjusted close-to-close cumulative NAV from the executed notebook; values include the modelled turnover cost for the strategy.

4. Calendar-year and partial-year robustness

Period	Strategy	SPY	Excess (pp)
2020 (May-Dec)	37.70%	30.35%	+7.35
2021	44.69%	28.73%	+15.96
2022	-26.81%	-18.18%	-8.63
2023	32.65%	26.18%	+6.48
2024	24.26%	24.89%	-0.62
2025 (Jan-Sep)	22.63%	14.66%	+7.97

The restricted strategy outperformed SPY in four of six displayed calendar segments, but underperformed materially in 2022, when its loss exceeded the benchmark's loss by 8.63 percentage points. Because 2020 and 2025 are partial years, the six rows should not be treated as six independent, full-year tests.

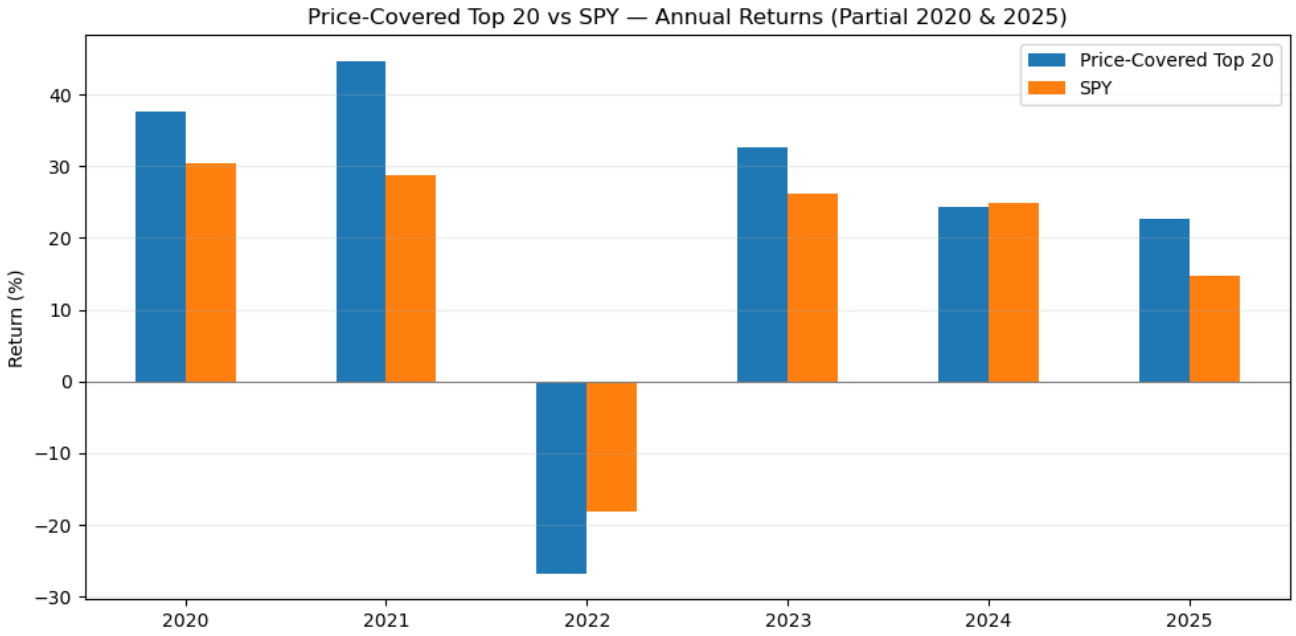


Figure 2. Calendar-year and partial-year returns for the Price-Covered Top 20 portfolio versus SPY.

5. Critical limitations and validity

• Point-in-time availability. Lagging weights by one trading day does not resolve uncertainty over when fundamentals became public. Some observed `reportdate` values precede their associated quarterly `date`; no release-date reconciliation was completed.

• Coverage and survivorship. Restricting ranking to symbols with supplied price columns removes 67 predicted tickers and changes the investment universe. Historical constituents and delisting proceeds are not fully audited; performance may be biased.

• Model/portfolio selection. Portfolio size and covered-universe implementation were assessed after inspecting performance over this historical window. The resulting figures are not independent out-of-sample evidence.

• Execution realism. Target-weight turnover is not adjusted for price-induced weight drift; fees omit spread, slippage, market impact and full initial entry. The observed average absolute-weight-change measure was 0.0455 per day.

• Price adjustments. Strategy prices derived from Compustat-style columns and SPY auto-adjusted prices may not share identical corporate-action or dividend conventions. Missing active returns were checked, but corporate-action completeness was not established.

6. Conclusion and next research steps

A sector-based machine-learning signal combined with a price-covered equal-weight Top-20 portfolio generated higher historical cumulative return than SPY during the evaluated period. The excess growth came with inferior risk-adjusted performance and substantially greater drawdowns. The result is best read as a documented experiment and a foundation for further research, not as verified investable alpha.

The next scientific steps are to establish actual financial-report publication timestamps, reconstruct a point-in-time tradable universe including delisted names and cashflows, harmonize adjusted-return definitions, model executed rebalancing costs, and lock the portfolio specification before a genuine walk-forward/out-of-sample evaluation. No such follow-up validation is claimed here.

Reproducibility and provenance

Primary source: user-executed `FinRL-Trading.ipynb`, including its saved numeric outputs and two embedded figures. Key local inputs referenced by the notebook: `prepared/daily.csv`, `prepared/final_ratios.csv`, sector workbooks, `outputs/stock_selection/stock_selected.csv`, and `outputs/backtest/spy_prices.csv`. The executable notebook and its companion code and datasets must be provided or documented separately for full replication. The PDF itself does not include those data files.